
LEARNABLE OBFUSCATION FOR TEMPORALLY RELATED VIDEO DATA

William Chang

Department of Applied Mathematics
University of California, Los Angeles
Los Angeles, CA
<https://williamc.me/>

ABSTRACT

1 Introduction

Modern machine learning systems increasingly rely on outsourced computation: data owners transmit datasets to remote service providers that perform model training and inference at scale. While this paradigm enables efficient deployment of sophisticated models, it also introduces substantial privacy risks when the underlying data is sensitive, as explored by Shan et al. [2018]. In particular, video data—ubiquitous in surveillance systems, autonomous driving platforms, teleconferencing services, and edge analytics—contains rich temporal and semantic information about individuals and environments. Even when the intended task is limited to high-level classification, transmitting raw video streams to an untrusted server may reveal identifiable faces, behaviors, or contextual cues.

A recent line of work, including Christiano et al. [2024], formalizes the problem of *learnable obfuscation*: can one transform data prior to outsourcing so that (i) the transformed data remains PAC-learnable for a target task, and (ii) the transformation provably limits information leakage about the original dataset? In this paradigm, the data owner applies an encoding mechanism M locally, producing an obfuscated representation $M(X)$, and only this encoded version is transmitted to an external service provider. The downstream learning algorithm—whether training or inference—is then performed exclusively on $M(X)$ rather than on the raw dataset X . The external party, which may be honest-but-curious or fully adversarial, observes only the encoded data and must operate within this restricted view.

In contrast to fully homomorphic encryption (FHE), which enables exact computation over encrypted data but incurs substantial computational overhead Gentry [2009], Gorantala et al. [2023], learnable obfuscation permits structured randomization—such as linear masking and additive noise—while maintaining statistical learnability of the target task. The objective is therefore twofold: first, to ensure that a predictor trained or evaluated on $M(X)$ achieves performance comparable to one trained on X ; and second, to guarantee that observing $M(X)$ does not enable an adversary to reliably recover sensitive information about X . Xiao et al. [2024] establish a formal connection between privacy and utility in this setting by bounding the adversary’s inference success solely as a function of the mutual information between the original dataset X and its encoded representation $M(X)$. In this framework, mutual information serves as a quantitative bridge: it simultaneously upper bounds the adversary’s posterior inference advantage and characterizes the extent to which the encoding preserves task-relevant structure.

Existing analyses, however, primarily consider static datasets composed of independent and identically distributed samples. While systems such as Chameleon exploit cross-camera correlations to adapt inference configurations and improve efficiency, they focus on accuracy–cost tradeoffs rather than formal privacy guarantees tied to information leakage Jiang et al. [2018]. Video data fundamentally departs from this assumption. Frames exhibit strong temporal dependence, events span contiguous segments, and semantic similarity between adjacent frames varies continuously rather than discretely. Indeed, modern video object detection systems explicitly exploit temporal coherence at the feature level, aggregating representations across motion trajectories to improve recognition accuracy Zhu et al. [2017]. Recent systems work further emphasizes that large camera deployments exhibit significant spatio-temporal correlations across views, which can be leveraged to improve efficiency and accuracy in video analytics Jain et al. [2019]. Consequently,

classical notions of membership inference—where an adversary attempts to determine whether a specific datapoint belongs to a dataset—do not fully capture realistic privacy risks in video analytics. Early systems such as PrivacyCam and the PICO framework from Chattopadhyay and Boulton [2007] demonstrated that selective encryption of regions of interest in video streams can provide practical, invertible privacy protection at the camera level; however, these approaches did not provide formal information-theoretic guarantees relating privacy leakage to downstream learning performance. More recently, Wu et al. [2021] proposes a securely reversible transformation for video streaming that preserves high downstream analytics accuracy while empirically suppressing sensitive attributes, and Kang et al. [2020] develops learned semantic indexing mechanisms to accelerate machine learning-based queries over large unstructured datasets, including video corpora. However, these approaches focus on system efficiency and empirical evaluation rather than providing formal information-theoretic guarantees that characterize adversarial inference success as a function of the mutual information between original and transformed video streams.

In practical video settings, adversarial success is rarely binary. An adversary may not reconstruct an exact frame, yet may still identify frames that are temporally or semantically close to a sensitive event. For example, mislocalizing a face by several frames in a continuous stream still reveals sensitive temporal information. Thus, privacy notions for video must accommodate *graded correctness* rather than purely exact reconstruction or membership recovery.

To address this gap, we introduce a *weighted inference framework* tailored to temporally correlated video data. Instead of assigning a reward solely for exact correctness, we define a similarity-based reward function over frames: exact matches receive full weight, proximate frames receive partial credit, and unrelated frames receive zero reward. This induces a generalized success metric that interpolates between strict membership inference and approximate event localization. We characterize the adversary’s *prior success probability* under random guessing in this weighted setting using multivariate hypergeometric models, thereby quantifying the baseline difficulty of the inference task before observing $M(X)$.

Building on the mutual information framework of Xiao et al. [2024], we extend their analysis to weighted video inference. Our results show that adversarial success probability under graded rewards is controlled by the mutual information $I(X; M(X))$, adjusted by a factor reflecting the increased prior success probability induced by partial credit. In particular, when relaxing correctness from integer-weight (exact-match) inference to half-integer or $1/m$ -resolution inference, the adversary’s baseline success strictly increases. To preserve an equivalent privacy guarantee, the obfuscation mechanism must reduce mutual information proportionally. For Gaussian masking mechanisms of the form

$$M(X) = XW + B,$$

we derive explicit calibration conditions on the noise variance required to offset the increase in prior success probability under fractional-weight inference.

To ground the theory in practice, we adopt a temporally local linear masking architecture inspired by privacy-preserving video analytics systems such as Cangialosi et al. [2022]. We evaluate classification accuracy and adversarial inference performance on real video datasets as functions of estimated mutual information and noise magnitude. Our empirical results validate the predicted privacy–utility tradeoff and demonstrate that mutual information serves as a predictive quantity governing adversarial advantage in temporally correlated data.

Contributions. This work makes the following contributions:

- We formalize a weighted inference model for video privacy that captures partial temporal and semantic alignment between frames.
- We derive closed-form expressions for adversarial prior success probability under discretized similarity weights.
- We extend mutual-information-based privacy bounds to graded inference settings and quantify the additional noise required to maintain equivalent guarantees.
- We empirically validate the resulting privacy–utility tradeoff in temporally structured video datasets.

By integrating formal learnable obfuscation guarantees with temporally aware inference models, this work advances the theoretical foundations of privacy-preserving outsourced learning from static i.i.d. datasets to realistic video streams.

2 Related Work

Our work builds on learnable obfuscation, membership inference, heuristic data transformations, and formal privacy guarantees. We situate our contribution within these lines of research.

2.1 Learnable Obfuscation and Property-Preserving Transformations

A recent line of work formalizes the notion of property-preserving data transformations under explicit learnability constraints. Xiao et al. [2024] initiate a formal study of *learnable obfuscation*, investigating whether data encodings can preserve PAC learnability while providing provable privacy guarantees. They show that non-trivial property-preserving transformations cannot achieve cryptographic IND-CPA security in the worst case, establishing a fundamental impossibility result. On the positive side, they develop information-theoretic tools under PAC Privacy Xiao and Devadas [2023] to quantify leakage for heuristic obfuscations such as matrix masking, data mixing, and permutation, and derive mutual-information-based bounds against reconstruction and membership inference attacks.

Parallel to this formal line of work, heuristic property-preserving transformations have appeared in several systems, including TransNet He et al. [2020], NeuraCrypt Yala et al. [2021], and Syfer Yala et al. [2022], which rely on random linear projections or random neural networks. Data mixing approaches such as Instahide Huang et al. [2020] and its variants Borgnia et al. [2021], Liu et al. [2020] apply convex combinations of samples to obscure individual data points. Matrix masking techniques have also been studied in privacy and statistical disclosure contexts Ding et al. [2020], Zhao and Chen [2018], often motivated by the Johnson–Lindenstrauss lemma Larsen and Nelson [2017] and locality-preserving embeddings Indyk et al. [1997].

However, several of these constructions have been shown to lack formal security guarantees. Carlini et al. [2021a,b] demonstrate reconstruction and identification attacks against instance encoding schemes such as Instahide and NeuraCrypt, highlighting the gap between empirical obfuscation and provable privacy.

Related ideas also arise in privacy-aware representation learning, where the goal is to construct feature mappings that retain task-relevant information while suppressing sensitive attributes. Adversarial representation learning frameworks Edwards and Storkey [2015], Louppe et al. [2017] formalize this tradeoff by training encoders to minimize predictive power about protected variables, while information bottleneck–based approaches Alemi et al. [2016] explicitly regularize mutual information between inputs and latent representations. More recently, formal analyses of invariant and minimal sufficient representations Achille and Soatto [2018] connect compression and invariance to generalization and privacy. Although these works are primarily motivated by fairness or robustness rather than outsourced learning, they share the central objective of constructing property-preserving transformations that discard extraneous information.

Our work extends the formal line of Xiao et al. [2024] by incorporating temporally correlated video data and graded similarity-based inference models, which require new prior success characterizations beyond binary correctness. Unlike prior representation-learning approaches, we analyze such transformations under explicit membership inference guarantees and derive prior-dependent noise calibration conditions within a formal learnable obfuscation framework.

2.2 PAC Privacy and Formal Learnable Obfuscation Foundations

Our analysis builds on recent information-theoretic approaches to instance-aware privacy, particularly the PAC-privacy framework and its formalization of learnable obfuscation. Xiao and Devadas introduced *PAC Privacy* as a distribution-dependent privacy notion that quantifies the posterior inference advantage of an adversary observing the output of an arbitrary data-processing mechanism Xiao and Devadas [2023]. Unlike differential privacy, which provides input-independent worst-case guarantees, PAC Privacy characterizes instance-level inference hardness and admits automatic Monte Carlo estimation for black-box mechanisms, enabling practical analysis of complex randomized pipelines.

Building on this framework, Xiao, Suh, and Devadas formalized *learnable obfuscation* and established both possibility and impossibility results for property-preserving encodings Xiao et al. [2024]. They show that non-trivial encodings supporting learning cannot in general achieve cryptographic indistinguishability, but can provide provable information-theoretic protection against membership and reconstruction attacks when combined with calibrated random perturbations. These results provide the formal basis for analyzing mechanisms such as matrix masking, data mixing, and permutation under explicit inference-hardness guarantees.

More broadly, PAC Privacy belongs to a line of work characterizing privacy leakage through information-theoretic dependence between inputs and released outputs. Mutual-information-based analyses show that controlling $I(X; A(X))$ limits overfitting, reconstruction risk, and inference advantage Russo and Zou [2016], Raginsky et al. [2016], Xu and Raginsky [2017]. Related notions such as maximal leakage and coupled-worlds privacy similarly quantify adversarial success via divergence-based dependence measures rather than worst-case indistinguishability Esposito et al. [2019], Bassily et al. [2013].

Connections between privacy guarantees and statistical learnability have also been studied in differentially private learning, where stability characterizes when learning remains feasible under privacy constraints Wang et al. [2016]. These results reinforce the central theme of learnable obfuscation: privacy-preserving transformations must control inference leakage while preserving task-relevant structure.

Our work extends this information-theoretic framework to temporally correlated video data and similarity-aware inference objectives. In contrast to prior analyses focusing on binary membership recovery in static i.i.d. datasets, we introduce weighted inference models capturing graded temporal and semantic alignment between frames and derive the corresponding prior-success and noise-calibration consequences within the PAC-privacy framework.

2.3 Membership Inference and Reconstruction Attacks

Membership inference attacks were first demonstrated in practice by Shokri et al. [2017]. Subsequent work strengthened theoretical understanding of reconstruction and inference hardness, including bounds on training data reconstruction Balle et al. [2022], Guo et al. [2022] and general analyses of membership inference advantage Carlini et al. [2022], Thudi et al. [2022] Hilprecht et al. [2019].

While most prior membership inference work evaluates adversarial success under binary criteria—i.e., whether a specific record belongs to the training set—related lines of research demonstrate that privacy leakage can occur at more granular or aggregate levels. Property inference attacks show that adversaries may recover dataset-level attributes without identifying specific instances Ganju et al. [2018], Melis et al. [2019], and reconstruction analyses quantify partial recovery risk beyond exact membership inclusion Balle et al. [2022], Guo et al. [2022]. These results suggest that inference risk is not inherently binary, motivating our graded, similarity-based formulation in which adversarial success is weighted according to partial alignment rather than exact recovery.

Beyond instance-level attacks, recent work has explored more powerful and adaptive membership inference strategies, including label-only attacks that operate without confidence scores Choquette-Choo et al. [2021], and auditing-style attacks that assess whether entire datasets or distributions were used during training Sablayrolles et al. [2019]. Other analyses characterize membership inference through hypothesis testing and likelihood-ratio formulations, providing a statistical perspective on adversarial advantage Yeom et al. [2018]. These developments further reinforce that inference risk extends beyond simple binary inclusion tests and can manifest through distributional, aggregate, or confidence-based signals, complementing the graded formulation adopted in our setting.

2.4 Differential Privacy and Its Limitations

Differential privacy (DP) provides worst-case indistinguishability guarantees Dwork [2006], Duchi et al. [2013], Bun and Steinke [2016], and has been extended to subsampling amplification Balle et al. [2018]. DP-based analysis has also been used to derive bounds on membership inference hardness Thudi et al. [2022] and reconstruction risk Balle et al. [2022].

While differential privacy (DP) provides strong worst-case indistinguishability guarantees that hold uniformly over all neighboring datasets Dwork [2008], Duchi et al. [2013], Bun and Steinke [2016], its input-independent nature can make tight, data-dependent analysis of complex transformations difficult. In contrast, information-theoretic leakage measures such as mutual information have been used to quantify inference advantage in a distribution-aware manner Russo and Zou [2016], Raginsky et al. [2016], Xiao and Devadas [2023]. Our framework adopts this instance-aware perspective, allowing us to directly characterize how relaxing correctness criteria increases adversarial prior success probability.

2.5 Cryptographic and Secure Computation Approaches

Cryptographic solutions such as Fully Homomorphic Encryption (FHE) provide computational security for outsourced learning Gentry [2009], Gorantala et al. [2023], but incur significant computational overhead. Other approaches include searchable encryption Kamara and Papamanthou [2013], Stefanov et al. [2013] and order-preserving or order-revealing encryption Agrawal et al. [2004], Popa et al. [2013], Lewi and Wu [2016], which enable limited functionality but do not preserve statistical learnability.

Xiao et al. [2024] show that non-trivial learnable obfuscation cannot achieve cryptographic indistinguishability in general. Our work operates within this information-theoretic regime, focusing on quantifying leakage rather than achieving computational secrecy.

Summary. Prior work studies heuristic property-preserving transformations, membership inference and reconstruction attacks, differential privacy, and cryptographic approaches to outsourced learning. We extend the formal learnable obfuscation framework to temporally structured video data, introducing graded similarity-based inference and deriving weight-dependent noise calibration rules.

3 Preliminary

We formalize a notion of membership inference closely related to Definition 7 of Xiao et al. [2024]. Membership inference is a fundamental and widely studied problem in data privacy Backes et al. [2016], Shokri et al. [2017]. In the standard setting, a dataset $X = \{x_1, \dots, x_n\}$ is sampled at random from a finite universe $U = \{u_1, u_2, \dots, u_N\}$, and the adversary’s goal is to determine which elements of U are contained in X . This objective can be evaluated either in an average-case sense (over the randomness of the dataset) or at the level of a particular individual instance.

Our setting departs from the classical formulation in an important way motivated by video inference. In video data, frames are not unrelated: neighboring frames are typically highly correlated. Consequently, if the learner aims to prevent the adversary from identifying whether a particular event or individual appears in frame f_i , then an adversary who predicts a nearby frame (e.g., f_{i-1}) should be considered partially successful rather than entirely incorrect. In other words, the learner seeks to prevent accurate localization, while allowing that “near misses” carry some informational value.

To capture this structure, we introduce a notion of *weights*. In the classical binary formulation, the adversary receives weight 1 if the event e occurs in frame f_i , and weight 0 otherwise, i.e.,

$$w_i = \begin{cases} 1 & \text{if } e \in f_i, \\ 0 & \text{if } e \notin f_i. \end{cases}$$

In contrast, in the video setting we allow the weight associated with a frame f_i to take values in the interval $[0, 1]$. Frames that are temporally close to one containing the event may receive intermediate weights, reflecting partial correctness due to temporal correlation. This weighted formulation generalizes the standard membership inference problem and better captures the structure inherent in sequential data.

Definition 1 (Membership Inference Challenge) *Let $U = \{u_1, u_2, \dots, u_N\}$ be a finite pool of video frames and let $\mathcal{M}$ be a (possibly randomized) processing mechanism. A dataset $X \subseteq U$ of size n is drawn uniformly at random from U , and the adversary observes the released output $\mathcal{M}(X)$. Based on this observation, the adversary outputs an n -subset $\hat{X} \subseteq U$ as its estimate of the true membership set X .*

We say that $\mathcal{M}$ satisfies a $(1 - \bar{\delta})$ average membership inference challenge if, for every adversary,

$$\mathbb{E}_X \left[\sum_{i=1}^n w_i \right] \leq n(1 - \bar{\delta}), \quad (1)$$

where $w_i \in [0, 1]$ denotes the weight assigned to the i th selected frame in $\hat{X}$ with respect to the true subset X . The expectation is taken over the random draw of X (and any randomness in $\mathcal{M}$). Thus, on average, no adversary can accumulate more than $n(1 - \bar{\delta})$ total weight from correctly (or approximately correctly) identifying elements of X .

The average guarantee above controls the adversary’s aggregate success across the entire dataset. However, such aggregate control is fundamentally determined by the adversary’s performance on individual elements. The individual guarantee, following Definition 7 of Xiao et al. [2024], formalizes this stronger, per-element notion of protection. In particular, if a uniform bound $\delta_i \geq \delta$ holds for all $u_i \in U$, then the average parameter satisfies $\bar{\delta} \geq \delta$. Conversely, an average guarantee alone does not preclude the possibility that certain elements are significantly more vulnerable than others. We therefore distinguish between aggregate protection and per-frame robustness.

Individual guarantee. For a fixed element $u_i \in U$, we say that $\mathcal{M}$ is resistant to $(1 - \delta_i)$ individual membership inference for u_i if, for every adversary,

$$\Pr(\mathbf{1}_{\{u_i \in X\}} = \mathbf{1}_{\{u_i \in \hat{X}\}}) \leq 1 - \delta_i,$$

where $\mathbf{1}_{\{u_i \in X\}}$ and $\mathbf{1}_{\{u_i \in \hat{X}\}}$ are indicator variables denoting whether u_i belongs to the true set X and the adversary’s estimate $\hat{X}$, respectively. This condition ensures that the adversary’s success probability in correctly determining the membership of any particular frame u_i is uniformly bounded away from 1.

In the classical (binary) membership inference setting, the weights satisfy $w_i \in \{0, 1\}$, so the left-hand side of (1) simply counts the number of correctly identified elements. In the video setting, however, we allow fractional weights $w_i \in [0, 1]$ to reflect temporal correlation between frames. If a frame returned by the adversary is close to a true frame in X , it may receive partial credit rather than being counted as entirely incorrect. As a result, the quantity on the left-hand side of (1) can only increase relative to the binary case. Consequently, satisfying the same upper bound $n(1 - \bar{\delta})$ is strictly more demanding in the weighted setting, since the adversary can accumulate success even from near-miss predictions. This captures the additional difficulty of ensuring privacy when the data exhibit strong temporal dependence.

4 Main Results

We now provide explicit bounds on the Inference Challenge in Definition 1 in terms of the mutual information between the original dataset and the released (or mixed) data. Mutual information serves as a quantitative measure of statistical dependence between two random variables. In our setting, it captures how much information about the sensitive dataset X is revealed by the mechanism's output $\mathcal{M}(X)$. A smaller mutual information implies that $\mathcal{M}(X)$ leaks less information about X , and consequently limits the power of any inference adversary.

To formalize this connection, we follow the general framework that relates posterior inference advantage to an f -divergence. Let Δ_f^δ denote the posterior advantage defined via a divergence $\mathcal{D}_f$. Then one can bound

$$\Delta_f^\delta = \mathcal{D}_f(\mathbf{1}_\delta \parallel \mathbf{1}_{\delta_0^\rho}) \leq \inf_{P_W} \mathcal{D}_f(P_{X, \mathcal{F}(X)} \parallel P_X \otimes P_W), \quad (2)$$

where $P_{X, \mathcal{F}(X)}$ is the joint distribution of $(X, \mathcal{F}(X))$, P_X is the marginal distribution of X , and P_W is an arbitrary distribution on the output space. This inequality shows that the adversary's inference advantage is controlled by how far the joint distribution is from a product distribution, i.e., from statistical independence.

A particularly important specialization is obtained by choosing $\mathcal{D}_f$ to be the Kullback–Leibler (KL) divergence and $P_W = P_{\mathcal{F}(X)}$. In this case, the right-hand side reduces to the mutual information between X and $\mathcal{F}(X)$, yielding

$$\Delta_{\text{KL}}^\delta = \mathcal{D}_{\text{KL}}(\mathbf{1}_\delta \parallel \mathbf{1}_{\delta_0^\rho}) \leq \text{MI}(X; \mathcal{F}(X)). \quad (3)$$

Here, $\text{MI}(X; \mathcal{F}(X))$ denotes the mutual information, and

$$\mathcal{D}_{\text{KL}}(\mathbf{1}_\delta \parallel \mathbf{1}_{\delta_0^\rho}) = \delta \log\left(\frac{\delta}{\delta_0^\rho}\right) + (1 - \delta) \log\left(\frac{1 - \delta}{1 - \delta_0^\rho}\right),$$

with $\log$ denoting the natural logarithm. Thus, (3) provides a direct and interpretable link between inference hardness and mutual information: any mechanism with small $\text{MI}(X; \mathcal{F}(X))$ necessarily limits the achievable posterior advantage of an adversary.

Moreover, mutual information admits explicit upper bounds under differential privacy guarantees. In particular, if $\mathcal{F}$ satisfies ε -differential privacy and X consists of n datapoints, then

$$\text{MI}(X; \mathcal{F}(X)) \leq \frac{1}{2} \varepsilon^2 n^2. \quad (4)$$

Combining (3) and (4) yields an explicit pathway from privacy parameters (such as ε) to concrete upper bounds on the adversary's success in the Membership Inference Challenge. In this way, mutual information acts as the central bridge connecting statistical dependence, differential privacy guarantees, and quantitative limits on membership inference performance.

Lemma 2 (Membership Inference Hardness) *For an arbitrary processing mechanism $\mathcal{M}$ and a normalized n -random-subset X from the N -universe U , the adversary's posterior success rate to identify the participation of some $u_i \in U$ is upper bounded by $(1 - \delta_i)$, where for sampling rate $q = n/N$ and*

$$\begin{aligned} 1 - \delta_0 &= \max\{q, 1 - q\}, \\ (1 - \delta_i) \log\left(\frac{1 - \delta_i}{1 - \delta_0}\right) + \delta_i \log\left(\frac{\delta_i}{\delta_0}\right) &\leq \text{MI}(1_{u_i}, \mathcal{M}(X)) \end{aligned} \quad (5)$$

Moreover, the average failure rate $\bar{\delta}$ in Definition 1 satisfies

$$1 - \bar{\delta} \leq \sum_{j=1}^n \frac{\text{MI}(X; \mathcal{M}(X)) + \log 2}{-\log(1 - \delta_0^{\geq j})}. \quad (6)$$

Proof: Equation (5) immediately follows from (3), given that with a random guessing on X , the probability that one can positively identify the participation of an individual is $q = n/N$. As for $\delta_{\geq j}$, let $\delta_{\geq j}$ represent the posterior chance that the adversary can correctly identify at least j memberships of selected X . Let $\delta_0^{\geq j}$ be the optimal *a priori* success rate before observing $\mathcal{M}(X)$. Given the normalized sampling strategy, where we randomly select n_0 samples from each class, it is not hard to verify that

$$1 - \delta_0^{\geq j} = \sum_{l=j}^n \sum_{\{p_{1:c}\}: \sum_{z=1}^c p_z = l} \prod_{z=1}^c \left(\binom{n_0}{p_z} \binom{N_0 - n_0}{n_0 - p_z} / \binom{N_0}{n_0} \right).$$

Now, by (2), we have

$$\begin{aligned} D_{\text{KL}}\left(1_{\delta_{\geq j}} \parallel 1_{\delta_0^{\geq j}}\right) &= \delta_{\geq j} \log\left(\frac{\delta_{\geq j}}{\delta_0^{\geq j}}\right) + (1 - \delta_{\geq j}) \log\left(\frac{1 - \delta_{\geq j}}{1 - \delta_0^{\geq j}}\right) \\ &\geq -(1 - \delta_{\geq j}) \log(1 - \delta_0^{\geq j}) - \log(2) - \delta_{\geq j} \log(\delta_0^{\geq j}), \end{aligned}$$

where we use the fact that

$$-\delta_{\geq j} \log(\delta_{\geq j}) - (1 - \delta_{\geq j}) \log(1 - \delta_{\geq j}) \leq \log(2).$$

Therefore, given $\delta_{\geq j} \log(\delta_0^{\geq j}) \leq 0$, combining with (2), we obtain

$$1 - \delta_{\geq j} \leq \frac{\text{MI}(X; \mathcal{M}(X)) + \log(2)}{-\log(1 - \delta_0^{\geq j})}. \quad (7)$$

Now, by Fubini's theorem on expectation, we know that the expected number of memberships that an adversary can recover in X equals

$$\sum_{j=1}^n (1 - \delta_{\geq j}),$$

which is upper bounded by (7). Substituting the expression of $\delta_0^{\geq j}$, the claim follows. $\square$

In Xiao et al. [2024] when the weights are in integer increments, they show

$$1 - \delta_0^{\geq j} = \sum_{l=j}^n \sum_{\{p_{1:c}\}: \sum_{z=1}^c p_z = l} \prod_{z=1}^c \left(\binom{n_0}{p_z} \binom{N_0 - n_0}{n_0 - p_z} / \binom{N_0}{n_0} \right). \quad (8)$$

Here $\{p_{1:c}\}_l$ represents all sets of non-negative integers $\{p_1, p_2, \dots, p_c\}$ such that

$$\sum_{z=1}^c p_z = l.$$

This characterization relies fundamentally on integer-valued weights and exact membership counts. When fractional weights are introduced, the combinatorial structure of the prior changes, and the corresponding success probability must be reformulated to account for weighted overlap rather than exact intersection size. In the following sections, we analyze several specific fractional-weight discretization schemes and derive analogous expressions and bounds for the prior success probability in each case.

4.1 Half-Integer Weights

We begin by analyzing the simplest nontrivial fractional-weight setting, in which similarity is discretized at resolution $\frac{1}{2}$. This case is the simplest extension of the binary formulation, allowing partial credit for near-miss predictions while preserving tractable combinatorics.

When the fractional weights are restricted to $\frac{1}{2}$, we obtain the following result. Specifically, we assume that the weights satisfy

$$w_i \in \left\{0, \frac{1}{2}, 1\right\},$$

so that exact members of X receive full weight 1, elements that are close to X (for example, temporally adjacent or semantically similar frames) receive partial weight $\frac{1}{2}$, and unrelated elements receive weight 0. Under this model, the adversary's weighted overlap captures both exact membership and coarse similarity, yielding a strictly richer notion of inference success than in the integer-weight case.

Under this half-integer weighting scheme, the adversary's prior success probability is determined by the distribution of the weighted overlap between X and a uniformly random guess $\hat{X}$. As in the integer-weight case, this distribution admits an explicit combinatorial characterization. The following lemma derives the corresponding closed-form expression for the prior success probability under half-integer weights.

Lemma 3 (Prior success probability under fuzzy intersection) *Let U be a universe of size N and let $X \subseteq U$ with $|X| = n$. Suppose elements of U are assigned weights in $\{0, \frac{1}{2}, 1\}$ relative to X , with n elements having weight 1, m elements having weight $\frac{1}{2}$, and the remaining $N - n - m$ elements having weight 0.*

Let $\hat{X}$ be drawn uniformly at random from all n -subsets of U , and define the weighted overlap

$$W(X, \hat{X}).$$

Then for any threshold $j \in \frac{1}{2}\mathbb{Z}$, the prior probability that the weighted overlap exceeds j is

$$1 - \delta_0^{\geq j} = \sum_{a=0}^n \sum_{b=0}^{n-a} \mathbf{1}[a + \frac{1}{2}b \geq j] \cdot \frac{\binom{n}{a} \binom{m}{b} \binom{N-n-m}{n-a-b}}{\binom{N}{n}}. \quad (9)$$

Proof: Partition the universe U into three disjoint subsets:

$$A := X, \quad B := P(X), \quad C := U \setminus (A \cup B),$$

where $|A| = n$, $|B| = m$, and $|C| = N - n - m$. Elements of A have weight 1, elements of B have weight $\frac{1}{2}$, and elements of C have weight 0.

For a uniformly random n -subset $\hat{X}$, define

$$a := |\hat{X} \cap A|, \quad b := |\hat{X} \cap B|.$$

Then $n - a - b = |\hat{X} \cap C|$, and the weighted overlap satisfies

$$W(X, \hat{X}) = a + \frac{1}{2}b.$$

Because $\hat{X}$ is sampled uniformly from all n -subsets of U , the vector

$$(a, b, n - a - b)$$

follows a multivariate hypergeometric distribution with population sizes $(n, m, N - n - m)$ and sample size n . Hence

$$\Pr[a, b] = \frac{\binom{n}{a} \binom{m}{b} \binom{N-n-m}{n-a-b}}{\binom{N}{n}}.$$

The event $\{W(X, \hat{X}) \geq j\}$ is equivalent to $\{a + \frac{1}{2}b \geq j\}$. Summing the above probability over all (a, b) satisfying this condition yields (9). $\square$

Remark. When $m = 0$, the fractional-weight class B is empty and $W(X, \hat{X}) = |\hat{X} \cap X|$. In this case, the double sum collapses to the standard hypergeometric expression for the integer-weight prior, showing that Lemma 3 strictly generalizes the classical setting.

This characterization demonstrates how fractional weights alter the combinatorial structure of the prior relative to the integer-weight case. In the following subsection, we extend this analysis to a more general $\frac{1}{m}$ resolution discretization, deriving the corresponding prior success probabilities under arbitrary fractional weight increments.

4.2 $\frac{1}{m}$ Resolution Weights

We now generalize the preceding analysis to a broader class of fractional-weight models in which similarity scores are discretized in increments of $1/m$. In this setting, each element of U is assigned a weight in

$$\{0, \frac{1}{m}, \frac{2}{m}, \dots, 1\},$$

with exact membership corresponding to weight 1 and fractional weights encoding graded similarity relative to X .

This framework strictly extends the half-integer case and yields a nested family of inference models parameterized by m . For each fixed m , the adversary's success is determined by the distribution of a discretized weighted overlap between X and a uniformly random n -subset $\hat{X}$. Despite the increased granularity of the weight structure, the underlying sampling mechanism remains combinatorial: the count vector of selected elements across weight classes follows a multivariate hypergeometric distribution. Consequently, the prior success probability admits an exact closed-form characterization analogous to the integer and half-integer cases.

The following lemma derives the corresponding expression for the prior success probability under arbitrary $1/m$ discretization.

Lemma 4 (Prior success probability under discretized fuzzy intersection) *Let U be a universe of size N and let $X \subseteq U$ with $|X| = n$. Suppose each element $u \in U$ is assigned a weight*

$$w(u; X) \in \{0, \frac{1}{m}, \frac{2}{m}, \dots, 1\},$$

with exact membership corresponding to weight 1. Let M_r denote the number of elements in U with weight r/m , so that $\sum_{r=0}^m M_r = N$ and $M_m = n$.

Let $\hat{X}$ be drawn uniformly at random from all n -subsets of U , and define the weighted overlap

$$W(X, \hat{X}) = \sum_{u \in \hat{X}} w(u; X).$$

Then for any threshold $j \in \{0, \frac{1}{m}, \dots, 1\}$,

$$1 - \delta_0^{\geq j} = \sum_{\substack{k_0, \dots, k_m \geq 0 \\ \sum_{r=0}^m k_r = n}} \mathbf{1} \left[\sum_{r=0}^m \frac{r}{m} k_r \geq j \right] \cdot \frac{\prod_{r=0}^m \binom{M_r}{k_r}}{\binom{N}{n}}. \quad (10)$$

Equivalently, writing $J := mj$, the indicator condition can be expressed as

$$\mathbf{1} \left[\sum_{r=0}^m r k_r \geq J \right].$$

Proof: Partition U into weight classes

$$A_r := \{u \in U : w(u; X) = r/m\}, \quad |A_r| = M_r,$$

for $r = 0, 1, \dots, m$. These sets form a disjoint partition of U , and in particular $A_m = X$.

For a uniformly random n -subset $\hat{X}$, define

$$k_r := |\hat{X} \cap A_r|, \quad r = 0, \dots, m,$$

so that $\sum_{r=0}^m k_r = n$. Then

$$W(X, \hat{X}) = \sum_{r=0}^m \frac{r}{m} k_r.$$

Because $\hat{X}$ is sampled uniformly from all n -subsets of U , the count vector

$$(k_0, \dots, k_m)$$

follows a multivariate hypergeometric distribution with category sizes $(M_0, \dots, M_m)$ and sample size n . Hence

$$\Pr[k_0, \dots, k_m] = \frac{\prod_{r=0}^m \binom{M_r}{k_r}}{\binom{N}{n}}$$

for all nonnegative vectors satisfying $\sum_{r=0}^m k_r = n$.

The event $\{W(X, \hat{X}) \geq j\}$ is equivalent to

$$\sum_{r=0}^m \frac{r}{m} k_r \geq j,$$

or equivalently $\sum_{r=0}^m r k_r \geq mj$. Summing the above probability over all admissible count vectors satisfying this condition yields (10). $\square$

5 Weight Dependent Noise in Data Mixing

In this section, we connect the weighted membership inference models developed above to concrete noise calibration rules for learnable obfuscation mechanisms. Our goal is to quantify how the required level of obfuscation changes when the adversary is permitted graded (fractional-weight) success rather than exact recovery.

We proceed in two steps. First, we recall an information-theoretic upper bound on $MI(X; M(X))$ for Gaussian matrix masking mechanisms. Second, we combine this bound with the weighted prior analysis to derive explicit conditions on the noise variance required to maintain a fixed inference guarantee under relaxed correctness criteria.

Recall the following result from Xiao et al. [2024] (Theorem 2).

Theorem 5 (Inference Regarding Entire Set X) When $W \in \mathbb{R}^{d_0 \times d}$ and $B \in \mathbb{R}^{n \times d}$ are two independent random matrices, where each entry of W is i.i.d. $\mathcal{N}(0, 1)$ and each column of B is i.i.d. according to a multivariate Gaussian distribution $\mathcal{N}(0, \Sigma_B)$ for some non-singular covariance matrix Σ_B , then for X consisting of n randomly generated data points,

$$\mathcal{ML}(X; XW+B) \leq \frac{d}{2} \min \left\{ \underbrace{\mathbb{E}_{X, X'} \left[\text{Tr} \left(\Sigma_{X, B}^{-1} \Sigma_{X', B} \right) - n \right]}_{(I)}, \underbrace{\log \det \left(\mathbb{E}_X [I_n + \Sigma_B^{-1} \Sigma_X] \right) - \mathbb{E}_X \left[\log \det (I_n + \Sigma_B^{-1} \Sigma_X) \right]}_{(II)} \right\}. \quad (11)$$

Here $\Sigma_X = XX^\top$, $\Sigma_{X, B} = XX^\top + \Sigma_B$, and X' is distributed independently and identically as X . The operator $\det(\cdot)$ denotes the determinant, and $\text{Tr}(\cdot)$ denotes the trace of a matrix.

Corollary 6 (Theorem 2 with isotropic i.i.d. noise $\Sigma_B = \sigma I_n$) Assume the setting of Theorem 2, but let each column of $B \in \mathbb{R}^{n \times d}$ be i.i.d. as $\mathcal{N}(0, \sigma I_n)$ (equivalently, the entries of B are i.i.d. $\mathcal{N}(0, \sigma)$, so $\Sigma_B = \sigma I_n$). Then, for $\Sigma_X = XX^\top$ and $\Sigma_{X, B} = XX^\top + \sigma I_n$ (and X' an i.i.d. copy of X),

$$MI(X; XW+B) \leq \frac{d}{2} \min \left\{ \underbrace{\mathbb{E}_{X, X'} \left[\text{Tr} \left(\Sigma_{X, B}^{-1} \Sigma_{X', B} \right) - n \right]}_{(I)}, \underbrace{\log \det \left(\mathbb{E}_X \left[I_n + \frac{1}{\sigma} \Sigma_X \right] \right) - \mathbb{E}_X \left[\log \det \left(I_n + \frac{1}{\sigma} \Sigma_X \right) \right]}_{(II)} \right\}.$$

$$\text{Tr}(\Sigma_{X, B}^{-1} \Sigma_{X', B}) = \text{Tr}((XX^\top + \sigma I_n)^{-1} (X'X'^\top + \sigma I_n)) = \text{Tr}(X'^\top (XX^\top + \sigma I_n)^{-1} X') + \sigma \text{Tr}((XX^\top + \sigma I_n)^{-1}).$$

Moreover,

$$\text{Tr}(\Sigma_{X, B}^{-1} \Sigma_{X', B}) - n = \text{Tr}((XX^\top + \sigma I_n)^{-1} (X'X'^\top - XX^\top)).$$

Calibrating noise for half-weight versus integer-weight inference. Theorem 5 and Corollary 6 provide an explicit upper bound on the mutual information between the original dataset and its masked representation under Gaussian data mixing. Since mutual information controls adversarial inference advantage via Lemma 2, these results furnish the analytical bridge between noise magnitude and membership inference hardness. We now specialize this connection to compare integer-weight and half-weight inference models.

We consider two adversarial inference models: an *integer-weight* model and a *half-integer-weight* model. In the latter, the adversary is rewarded even when its guess is not exactly correct (e.g., selecting a nearby frame in a video), which induces a strictly larger prior success probability. As a result, for each $j \in [n]$, the prior success term satisfies

$$\delta_{0, \text{half}}^{\geq j} \geq \delta_{0, \text{int}}^{\geq j}.$$

Our goal is to determine how much additional noise must be added in the half-weight setting so that the adversary's overall probability of success remains no larger than in the integer-weight setting.

To formalize this comparison, we express the adversary's overall success probability in terms of the mutual information and the prior success terms derived in Section 4. This decomposition makes explicit how relaxing the weight structure affects the required privacy budget.

Adversary success bound. From Lemma 2, the adversary's success probability satisfies

$$1 - \bar{\delta} \leq \sum_{j=1}^n \frac{MI(X; \mathcal{M}(X)) + \log 2}{-\log(1 - \delta_0^{\geq j})}. \quad (12)$$

Define

$$D_j := -\log(1 - \delta_0^{\geq j}), \quad C := \sum_{j=1}^n \frac{1}{D_j}.$$

Since the mutual information term does not depend on j , the bound (12) can be rewritten as

$$1 - \bar{\delta} \leq (MI(X; \mathcal{M}(X)) + \log 2) C. \quad (13)$$

We next examine how the prior success terms differ between the integer-weight and half-weight models. Since fractional credit enlarges the adversary's feasible success region, the corresponding prior quantities necessarily change.

Effect of half weights. Because $\delta_{0,\text{half}}^{\geq j} \geq \delta_{0,\text{int}}^{\geq j}$, we have

$$D_{j,\text{half}} \leq D_{j,\text{int}} \Rightarrow C_{\text{half}} \geq C_{\text{int}}.$$

Thus, holding the mechanism $\mathcal{M}$ fixed, the bound on adversarial success is strictly worse under half-weight inference.

Noise calibration criterion. Let MI_{int} and MI_{half} denote the mutual information terms induced by the mechanism under the integer-weight and half-weight settings, respectively. A sufficient condition to ensure that the adversary’s success bound under half weights does not exceed that of the integer-weight case is

$$(\text{MI}_{\text{half}} + \log 2) C_{\text{half}} \leq (\text{MI}_{\text{int}} + \log 2) C_{\text{int}}. \quad (14)$$

Equivalently,

$$\text{MI}_{\text{half}} \leq \left(\text{MI}_{\text{int}} + \log 2 \right) \frac{C_{\text{int}}}{C_{\text{half}}} - \log 2. \quad (15)$$

Equation (15) identifies the mutual-information threshold necessary to counterbalance the adversary’s increased prior advantage under half-weight inference. We next convert this condition into an explicit requirement on the Gaussian noise variance in the masking mechanism.

Relating mutual information to Gaussian noise. In our mechanism $\mathcal{M}(X) = XW + B$, where W has i.i.d. $\mathcal{N}(0, 1)$ entries and B has i.i.d. Gaussian entries with variance σ , Theorem 2 yields

$$\text{MI}(X; XW + B) \leq \frac{d}{2} \min\{(I_\sigma), (II_\sigma)\},$$

where (I_σ) and (II_σ) are decreasing functions of the noise variance σ . Therefore, a sufficient choice of noise variance σ_{half} for the half-weight setting is any value satisfying

$$\frac{d}{2} \min\{(I_{\sigma_{\text{half}}}), (II_{\sigma_{\text{half}}})\} \leq \left(\text{MI}_{\text{int}} + \log 2 \right) \frac{C_{\text{int}}}{C_{\text{half}}} - \log 2. \quad (16)$$

Interpretation. Equation (16) shows that the additional noise required in the half-weight setting scales directly with the ratio $C_{\text{half}}/C_{\text{int}}$, which quantifies the increase in the adversary’s prior success due to relaxed correctness requirements. Thus, half-weight inference can be made no more powerful than integer-weight inference by increasing the Gaussian noise variance until the corresponding mutual information upper bound is reduced by this same factor.

6 Experiments

The preceding sections establish information-theoretic bounds that relate adversarial inference success to the mutual information $\text{MI}(X; M(X))$ between the original dataset and its obfuscated representation. While these guarantees provide formal control over privacy leakage, it remains essential to evaluate whether such bounds translate into meaningful and practical privacy–utility tradeoffs. In particular, our theory predicts that reducing $\text{MI}(X; M(X))$ limits adversarial success under both exact and weighted membership inference. However, aggressive reduction of mutual information may also distort task-relevant structure, especially in complex, high-dimensional data such as video.

Compared to static images, video data presents additional challenges: frames are temporally correlated, events span contiguous segments, and semantic content evolves continuously over time. As a result, obfuscation mechanisms must preserve not only spatial features but also temporal structure sufficient for downstream recognition. The privacy–utility tradeoff in video is therefore inherently more delicate than in the i.i.d. image setting. Prior work by Xiao et al. [2024] demonstrated that linear masking mechanisms of the form $M(X) = XW + B$ can achieve favorable privacy–utility tradeoffs for image classification tasks. Our experiments extend this paradigm to temporally structured video data, where both the mutual information and the adversarial inference objective are more nuanced due to graded, similarity-aware membership criteria.

To investigate these issues, we evaluate learnable obfuscation under the linear transformation

$$M(X) = XW + B,$$

where W is a random masking matrix and B is additive noise. By varying the distributional parameters governing W and B , we systematically modulate the mutual information between X and $M(X)$ and observe the resulting impact on downstream classification accuracy. This setup allows us to test whether mutual information continues to serve as a predictive and practically meaningful quantity governing the privacy–utility tradeoff in temporally correlated video data.

Our experiments are designed to validate that nontrivial operating regimes exist in which utility can be retained while the mechanism induces an information bottleneck consistent with our mutual-information-based analysis. The results will provide empirical evidence that information-theoretic privacy guarantees remain applicable in realistic video learning settings, despite the added structural complexity of sequential data.

Scope of this study. In the experiments reported below, we focus on the *utility* axis of the privacy–utility tradeoff by measuring downstream top-1 classification accuracy. While our theory also bounds adversarial success under weighted membership inference as a function of $\text{MI}(X; M(X))$, a direct empirical evaluation of weighted inference attacks is deferred to future work.

6.1 Experimental Setup

Goal and experimental question. Our theoretical results bound adversarial inference success in terms of $\text{MI}(X; M(X))$, and predict a monotone privacy–utility tradeoff as the obfuscation mechanism suppresses more information about the original data. The purpose of the experiments is therefore twofold: (i) to verify that a simple linear learnable obfuscation pipeline remains *useful* for video classification, and (ii) to establish an empirical operating regime in which utility is retained while the mechanism introduces an information bottleneck consistent with our mutual-information-based analysis.

Video embeddings. Directly applying masking to raw video pixels can confound utility and privacy effects with low-level image distortions and heavy compute. Instead, we follow the common systems and representation-learning approach of first extracting compact clip-level features and then applying obfuscation on those representations. This isolates the privacy–utility impact of the obfuscation mechanism itself. We evaluate on UCF-101 split 1, a standard action recognition benchmark with 101 classes, where frames are temporally correlated even though clips are trimmed around an action.

Pipeline overview. The full experiment consists of:

1. **Clip construction:** each video is converted into a fixed-length clip of $T = 16$ frames sampled across the full video, resized to 112×112 .
2. **Feature extraction:** each clip is embedded into a fixed-dimensional vector using a pretrained R3D-18 video model (Kinetics-400 weights), producing $d_0 = 512$ dimensional embeddings.
3. **Balanced training subset:** to match the normalized sampling strategy used in prior learnable obfuscation experiments, we train on a stratified subset of exactly 50 clips per class ($n = 5050$ total).
4. **Obfuscation (mixing + masking):** we apply class- k mixing to form a larger mixed dataset and then apply randomized linear masking of the form $M(X) = XW + B$ with additive Gaussian noise and permutations.
5. **Downstream learning:** we train a lightweight MLP classifier on either the clean embeddings (baseline) or the obfuscated embeddings (obfuscated setting), and evaluate top-1 test accuracy on the UCF-101 test split.

6.2 Mechanism Details and Training Configuration

Baseline (no obfuscation). For the baseline, we train an MLP classifier directly on the $d_0 = 512$ clip embeddings with standard cross-entropy loss using the balanced subset ($n = 5050$).

Obfuscation: class- k mixing, projection, and noise. For the obfuscated setting, we first generate a mixed dataset using class- k mixing: for each ordered class pair $(i, j) \in [c] \times [c]$ with $c = 101$, we sample k examples from class i and k from class j and average their embeddings to form one mixed example. This yields $m = c^2 = 10201$ mixed samples. When $i = j$, the target is one-hot; when $i \neq j$, the target is a soft label with weight $1/2$ on each of the two classes.

We then apply a randomized linear masking mechanism:

$$M(X_{\text{mix}}) = \Pi_1 X_{\text{mix}} W + B,$$

where $W \in \mathbb{R}^{512 \times 256}$ is a random Gaussian projection (with entries scaled by $1/\sqrt{d}$) (so the obfuscated dimension is $d = 256$), Π_1 is a random permutation over samples, and $B \in \mathbb{R}^{m \times 256}$ is i.i.d. Gaussian noise. In our implementation, the command-line parameter `-sigma` denotes the *noise variance*, so $B_{ij} \sim \mathcal{N}(0, \sigma)$ (equivalently, $\Sigma_B = \sigma I$). Following the label protection step in prior learnable obfuscation pipelines, we also permute label coordinates by a random permutation Π_2 (and invert it at evaluation).

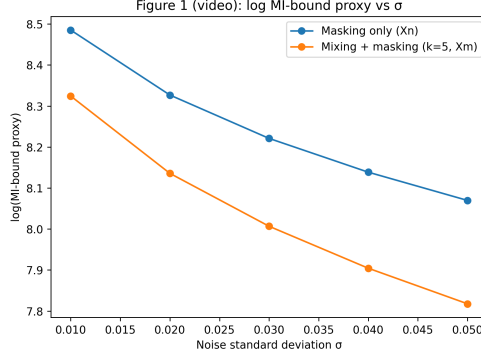


Figure 1: Figure 1 (video): Log MI-bound proxy vs. noise scale σ . We compute a log-det mutual-information upper-bound proxy on UCF-101 clip embeddings (R3D-18, $d_0 = 512$), comparing masking-only on the balanced subset X_n ($n = 5050$) to mixing+masking on the mixed set X_m ($m = c^2 = 10201$, $k = 5$). Each point is averaged over Monte Carlo trials with different random seeds.

Figure 1 replicates the role of Figure 1 in Xiao et al. [2024] for video embeddings: it shows how increasing the Gaussian noise scale σ monotonically decreases an MI-style upper-bound proxy. In our implementation, this MI-bound proxy is computed directly from the embedding matrix via a log-det expression as a function of σ , rather than by explicitly sampling W and B inside the proxy computation. We also observe that class- k mixing reduces the proxy bound relative to masking alone, consistent with mixing further suppressing dependence between X and $M(X)$.

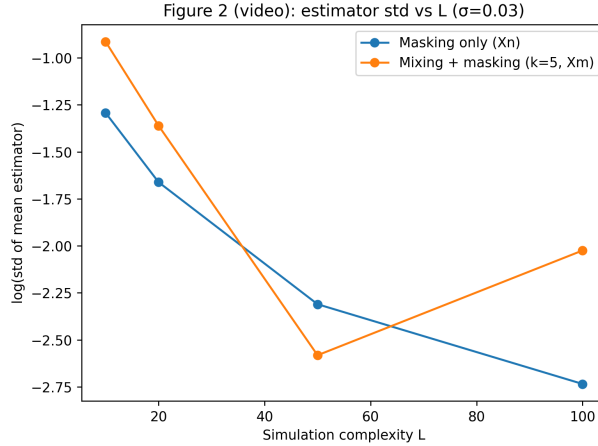


Figure 2: Figure 2 (video): Estimator variability vs. simulation complexity L at $\sigma = 0.03$. For fixed noise level $\sigma = 0.03$, we estimate the variability of the Monte-Carlo mean MI-bound proxy as the number of Monte-Carlo trials L increases, for masking-only on X_n and mixing+masking on X_m ($k = 5$). We plot $\log(\text{std}(\hat{\mu}_L))$, where $\hat{\mu}_L$ is the empirical mean of the MI-bound proxy over L trials.

Figure 2 replicates the role of Figure 2 in Xiao et al. [2024] by quantifying the stability of our Monte-Carlo estimator for the MI-bound proxy. As L increases, the standard deviation of the estimated mean generally decreases, indicating improved estimator reliability with more simulation trials. The mild non-monotonicity at larger L for the mixing+masking curve is expected under a finite number of repeats and reflects Monte-Carlo noise rather than a failure of convergence; increasing the number of repeats would smooth this effect. Overall, the trend supports that a moderate L suffices to obtain stable MI-proxy estimates for our video embedding pipeline.

Training protocol. We train the same MLP architecture in both settings (adjusting input dimension), and report top-1 test accuracy. Since mixing produces soft labels, the obfuscated model is trained using cross-entropy against soft targets. Unless otherwise noted, we use the same optimizer and learning rate across settings, and vary only the obfuscation pipeline (mixing + masking) and the epoch budget.

Setting	Epochs	n (train)	m (mixed)	Test Acc. (%)	Acc. Gap (%)
Baseline (no obfuscation)	10	5050	–	62.04	–
Obfuscated ($d = 256$, mixing + masking)	10	5050	10201	65.53	-3.49
Baseline (no obfuscation)	100	5050	–	73.46	–
Obfuscated ($d = 256$, mixing + masking)	100	5050	10201	71.95	1.51

Table 1: UCF-101 split 1 results using cached clip embeddings ($d_0 = 512$) and a balanced training subset of 50 clips per class ($n = 5050$). Obfuscation uses class- k mixing to construct $m = c^2 = 10201$ mixed samples, followed by randomized projection to $d = 256$ with additive Gaussian noise and permutations. Acc. Gap is $\text{baseline} - \text{obfuscated}$ (negative means obfuscated is higher).

Fixed obfuscation parameters. For the results in Table 1, we hold the masking distribution fixed and use the default parameters, including projection dimension $d = 256$ and noise level $\sigma = 0.03$. We then vary only the training budget (10 vs. 100 epochs) to illustrate how the obfuscation bottleneck affects early versus long-horizon generalization.

Table 1 reports our main UCF-101 split 1 results under two training budgets. With a short budget (10 epochs), the obfuscated model slightly outperforms the baseline. With a longer budget (100 epochs), the baseline improves more than the obfuscated model, producing a modest baseline advantage.

Interpretation and connection to our theory. These results are consistent with the qualitative behavior implied by our framework. Mixing and additive noise introduce an information bottleneck that reduces dependence between X and $M(X)$, which is the quantity controlling adversarial advantage in our bounds. Empirically, this bottleneck can act as an implicit regularizer: at small epoch budgets, the obfuscated learner can generalize competitively (and even outperform the baseline). However, as optimization proceeds, the baseline continues to benefit from access to higher-fidelity representations, while the obfuscated learner can saturate earlier due to the projection/noise constraints. This illustrates the central privacy–utility tension in temporally structured video settings: privacy mechanisms that suppress leakage can preserve strong *initial* utility while limiting asymptotic accuracy under longer training.

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